Performance-Based Design

CUNY EES 79903

Seminar on Nature-Based Solutions

Kyle McKay

November 10, 2026

Lecture Overview:

- 415-420 – Housekeeping
- 420-445 – Discussion with Anna Hochhalter and Nicole Del Monaco
- 445-615 – Discussion: Ostrow and Cox (2026), Godsey et al. (2026)

	Leader #1	Leader #2
~~Sep 8: NBS fundamentals~~	~~Grace~~	~~Jesse~~
~~Sep 15: NBS fundamentals~~	~~Michelle~~	~~Shahela~~
~~Sep 22: Structured decision making and community engagement~~	~~Jesse~~	~~Aurelia~~
~~Sep 29: Analysis – ecological~~	~~Carol~~	~~Julia~~
~~Oct 6: Analysis – co-benefits~~	~~Shahela~~	~~Aurelia~~
~~Oct 20: Analysis – engr performance~~	~~Grace~~	~~Michelle~~
~~Oct 27: Analysis – engr performance~~	~~Abdellah~~	~~Julia~~
~~Nov 3: Alternatives analysis~~	~~Aurelia~~	~~Abdellah~~
Nov 10: Performance-based design	Aaryan	Michelle
Nov 17: Constructability	Grace	Aaryan
Nov 24: Performance monitoring	Shahela	Jesse
Dec 1: Performance monitoring	Julia	Carol
Dec 8: Life cycle analysis	Aaryan	Abdellah
Dec 15: Course synthesis	Kyle	Brett

Project Reminder

- ~~Oct 20 – Recommended timeframe to compile initial references~~
- ~~Nov 3 – Recommended timeframe for outlining ideas~~
- Nov 17 – Initial submission of traceable account
- Nov 24 – Review of traceable account by classmates
- Dec 8 – Final submission of traceable account
- Dec 15 – Course synthesis (McKay)

Topic	First option
Qualitative foundations of project planning	Jesse
Facilitation, engagement, and teaming	Aurelia
Ecological analysis of NBS	Julia
Economic analysis of NBS	
Co-benefits analysis of NBS	
Engineering analysis of NBS	Abdellah
Alternatives analysis and structured decision making	
Performance-based design	Aaryan
Constructability	Michelle
Performance monitoring	
Project establishment	
Asset management	Grace

Paper Discussion

Godsey, E.S., Everett, C.L., Tritinger, A.S., Bryant, D.B., Altman, S., Sowers, A.A., Bassetti, L., Bruck, J. and Bonura, C., 2026. Engineering practice guide for oyster reefs: a natural infrastructure approach for coastal systems.

Ostrow, K.S. and Cox, D.T., 2026. Performance-Based Design Methodology for Nature-Based Solutions: Using Mangroves to Mitigate Wave Overtopping. ASCE OPEN: Multidisciplinary Journal of Civil Engineering, 4(1), p.04026006.

Failure Criterion

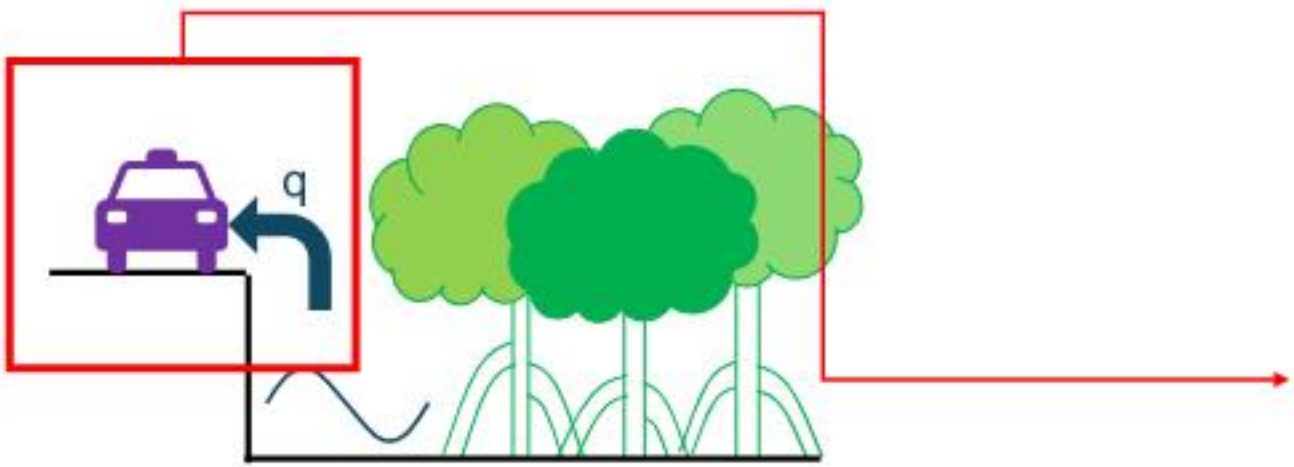
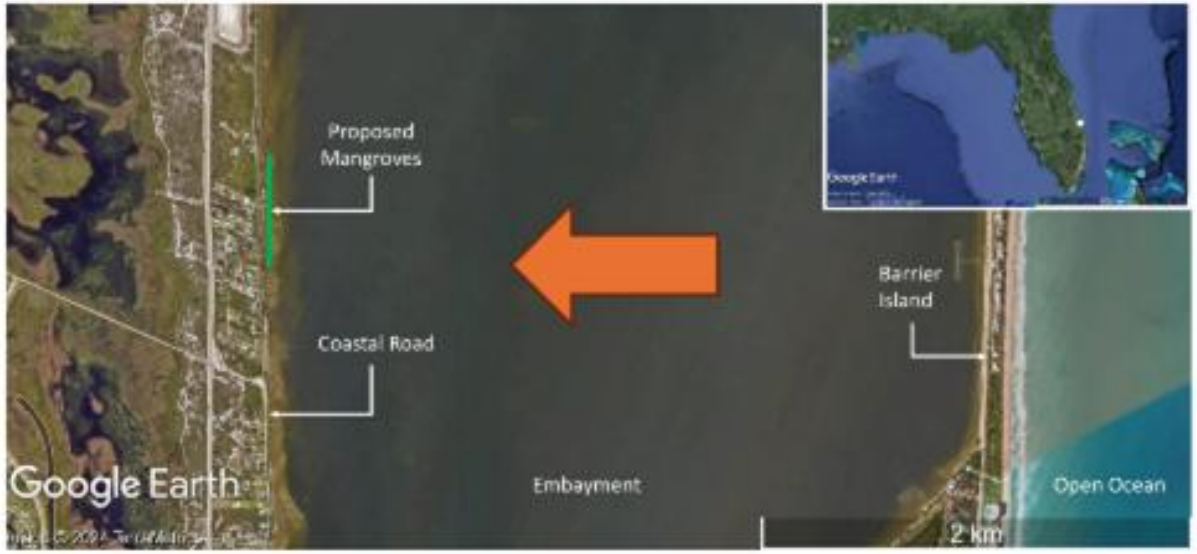


Table VI-5-6
Critical Values of Average Overtopping Discharges

q m³/s per m	SAFETY OF TRAFFIC		STRUCTURAL SAFETY				q litres/s per m
	VEHICLES	PEDESTRIANS	BUILDINGS	EMBANKMENT SEAWALLS	GRASS SEA-DIKES	REVETMENTS	
10^{0}						Damage even for paved promenade	1000
				Damage even if fully protected			200
10^{-1}					Damage	Damage if promenade not paved	100
	Unsafe at any speed	Very dangerous	Structural damage	Damage if back slope not protected			50
10^{-2}				Damage if crest not protected			20
					Start of damage		10
10^{-3}							2
			Dangerous on grass sea dikes, and horizontal composite breakwaters				1
	Unsafe parking on horizontal composit breakwaters	Dangerous on vertical wall breakwaters				No damage	
10^{-4}	Unsafe parking on vertical wall breakwaters						0.1
				No damage			0.03
		Uncomfortable but not dangerous					0.02
10^{-5}			Minor damage to fittings, sign posts, etc.		No damage		0.01
	Unsafe driving at high speed						0.004
10^{-6}							0.001
		Wet, but not uncomfortable					
	Safe driving at all speeds		No damage				0.0001
10^{-7}							

Figure from Kayla Ostrow (July 2025)

Idealized case study



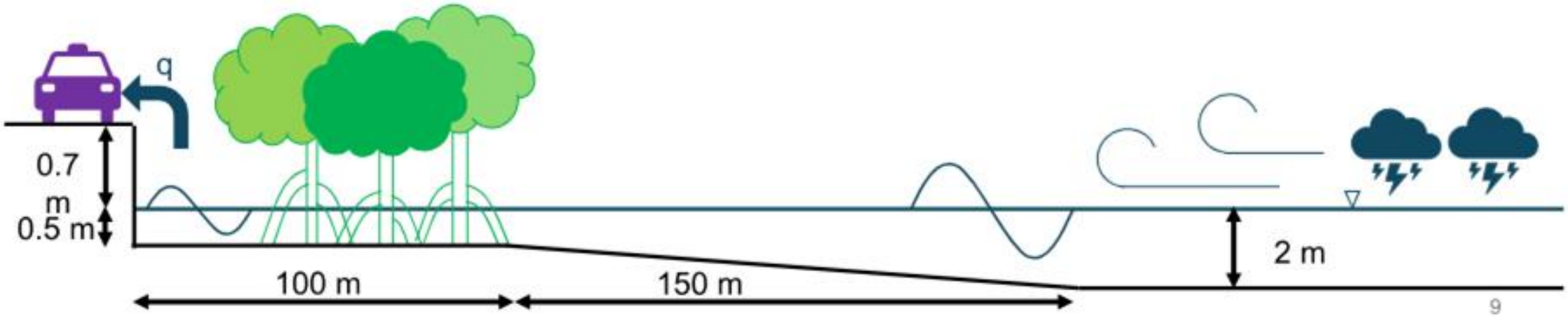
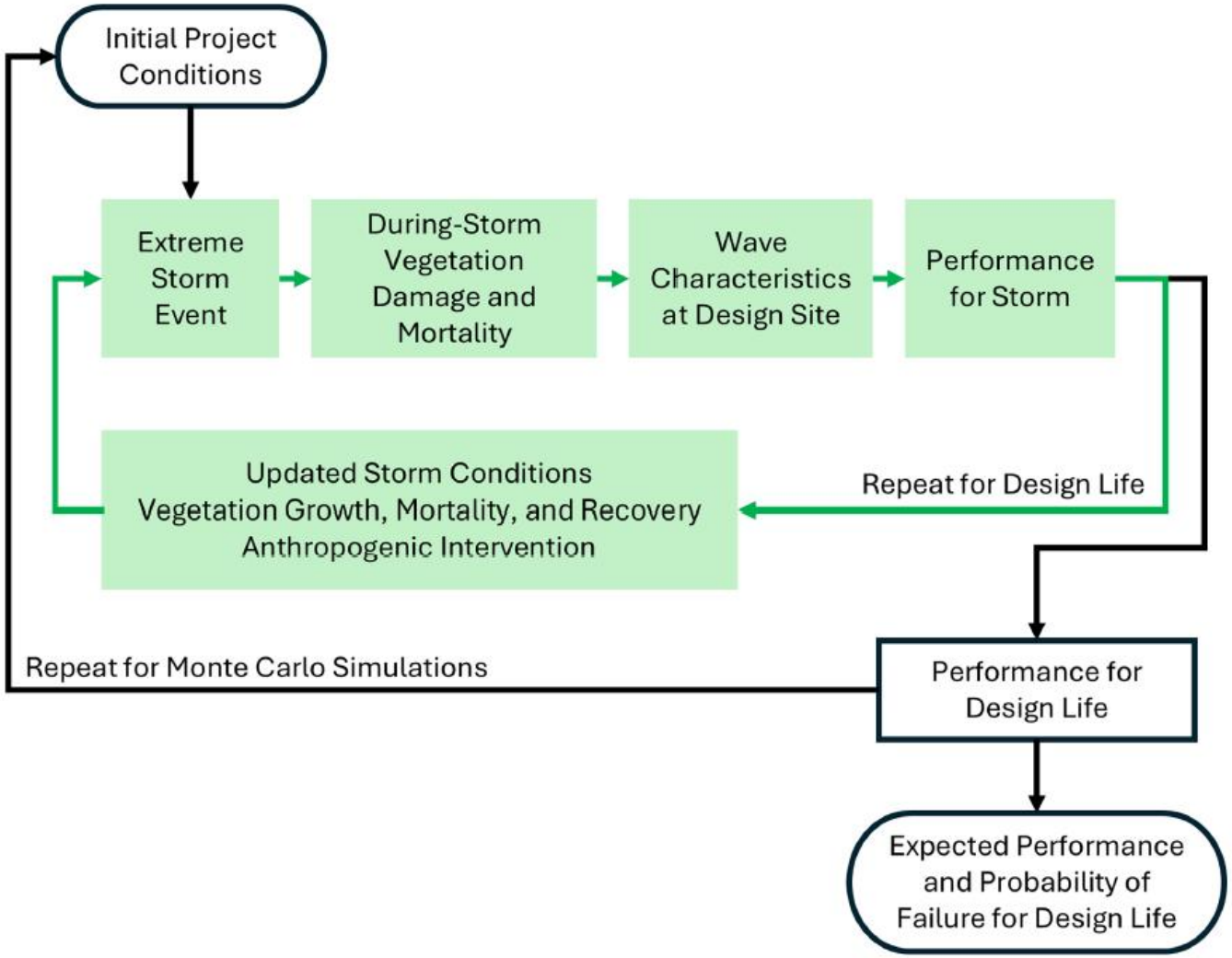


Figure from Kayla Ostrow (July 2025)

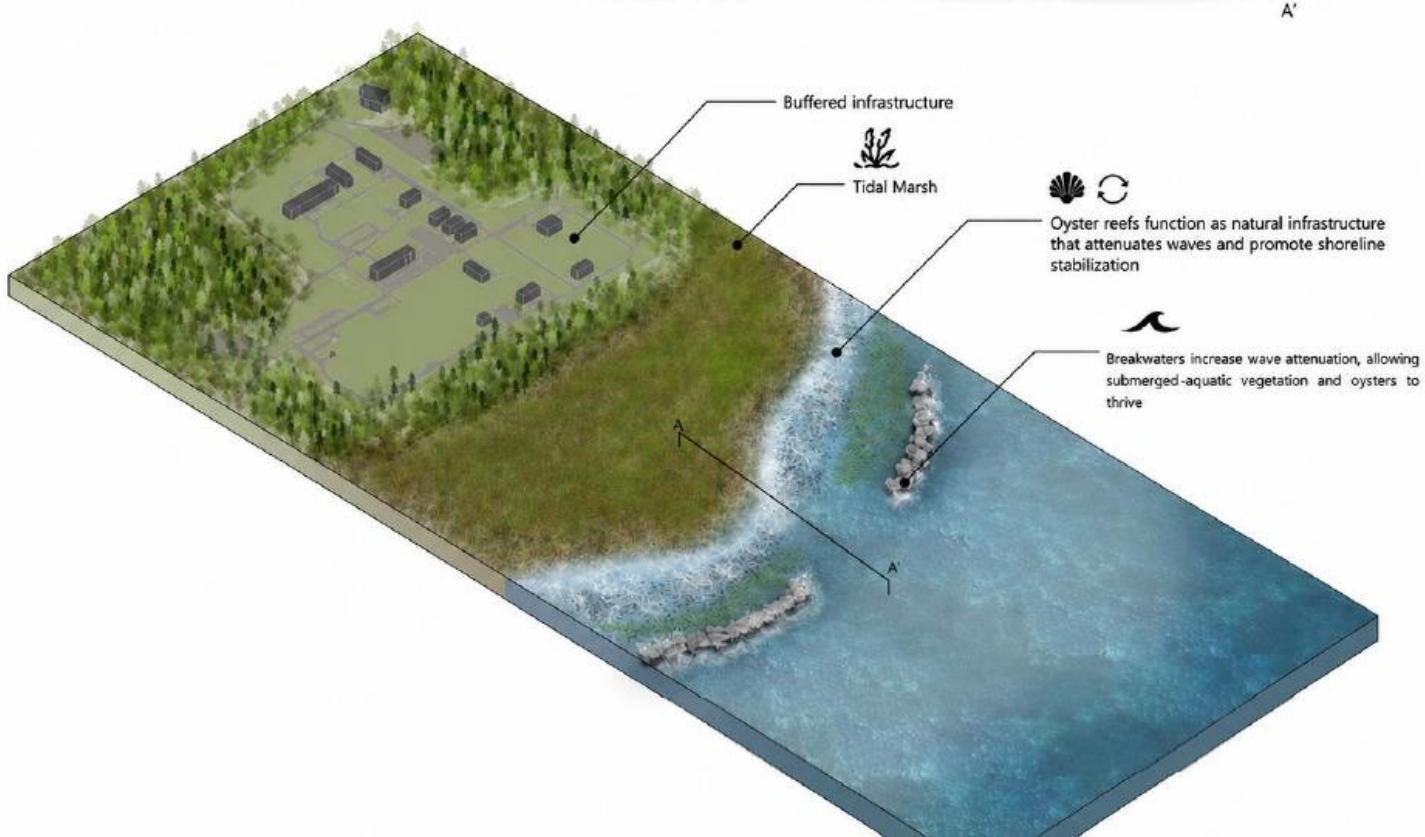
Ostrow and Cox (2026)

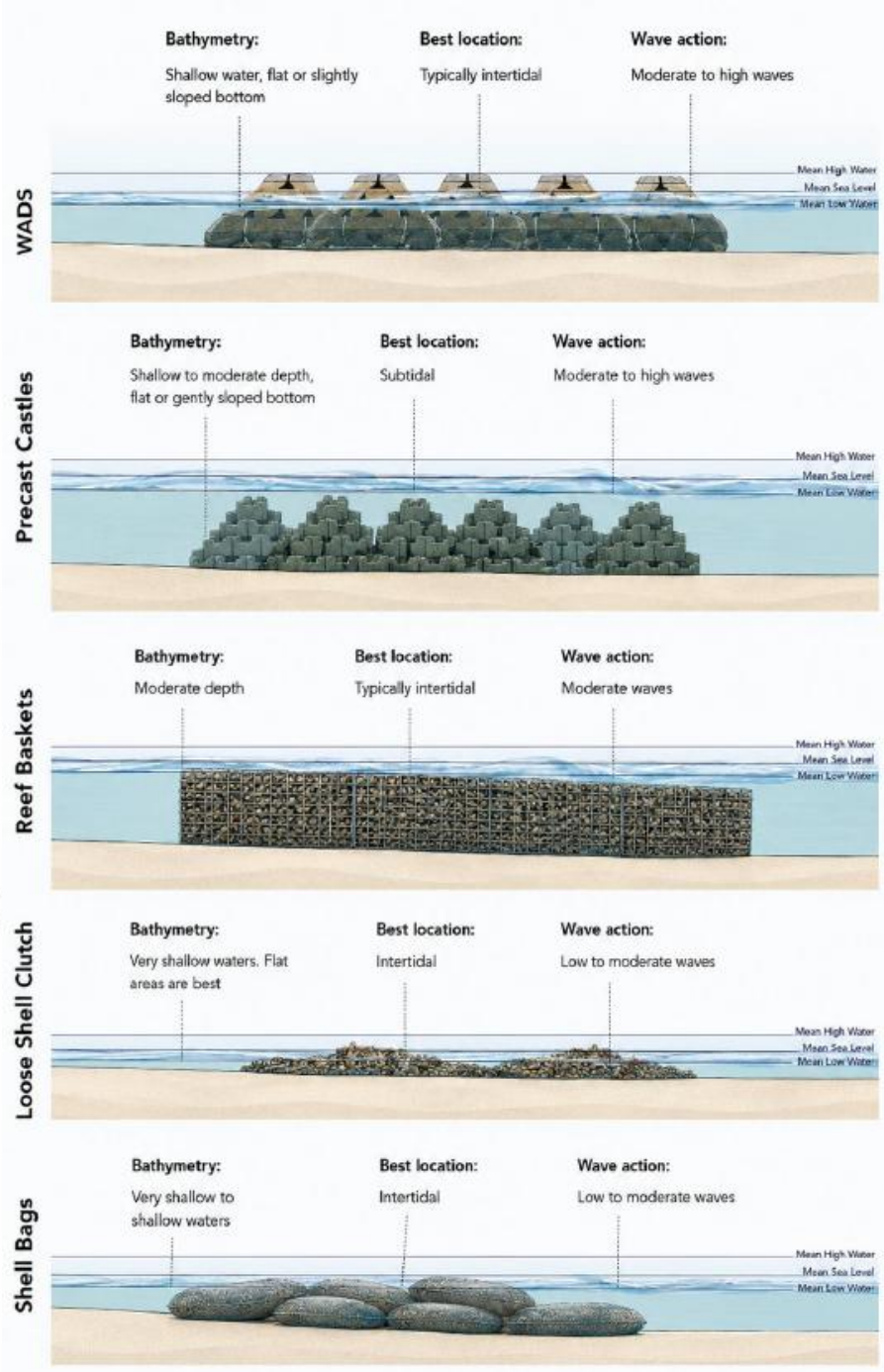
- The design process seems simple, so why are the methods so complex?
- Where does performance-based design fit on a spectrum of qualitative to quantitative design approaches?
- How would this process change for conventional infrastructure? What about other types of NBS?



Godsey et al. (2026)







Godsey et al. (2026)

- Site assessment: How much information do you need to execute a robust design? How do you know when to stop gathering data?
- Design: How do you explore the breadth of so many design parameters?

Table 19. Functional design trade-offs.

Design Factor	Ecological Optimization	CSRM Optimization	Hybrid Compromise
Orientation	Perpendicular to tidal flow for larval settlement	Parallel to shore for wave attenuation	Angled or segmented to balance flow and shoreline alignment
Crest Elevation	Intertidal (mean low water to mean high water) or subtidal for oyster survival	At or above mean high water for wave energy reduction	Mixed intertidal/subtidal to support oysters and reduce wave energy with careful consideration of impacts to shoreline response
Crest Width	Narrow crests enhance vertical relief and exposure; wide crests increase habitat area, stability, and biodiversity	Wide crests improve wave dissipation, structural mass, and durability; narrow crests may be used where space or material is limited	Variable crest widths tailored to site conditions; may combine narrow sections for ecological niches with wider sections for wave attenuation and structural integrity
Gap Spacing	Wider gaps for water flow and habitat diversity	Narrow or no gaps for continuous protection	Moderate gaps to allow flow while maintaining structural continuity
Distance from Shore	Closer for habitat connectivity	Farther for wave dissipation before shoreline impact	Intermediate distance to balance ecological access and wave attenuation

Table 11. Relevant design parameters for oyster reefs.

Design Parameter	Description	Additional Information
Reef Length (L_R)	Length of continuous reef section (alongshore)	Patch size, shape, and spatial configuration may impact oyster recruitment and survival (Morris et al. 2019). Longer reef segments provide broader protection but increase material and installation costs.
Crest Width (B)	Cross-shore width of reef crest	Wave transmission decreases for wider crests (van Gent et al. 2023).
Crest Level or Freeboard (R_c)	Depth of still water over crest, expressed as freeboard (negative values refer to a submerged reef)	Freeboard is an important consideration for both wave transmission and ecological constraints. Oyster feeding and respiration requirements dictate the ideal tidal exposure ranges for habitat creation (see Section 2; Morris et al. 2019). However, wave transmission is also highly impacted by freeboard—the higher the crest elevation, the lower the wave transmission. Over time there may be changes to wave breaking and setup due to oyster colonization (Morris et al. 2019).
Depth of Reef (d_R)	Average water depth to seafloor at centerline or toe of reef, depending on formulation	Closely related to distance from shore, dependent on beach slope.
Distance from Shore (x_R)	Distance from centerline of reef to undisturbed shoreline	Closely related to depth of reef, dependent on beach slope. Placement relative to the shoreline determines the effectiveness of wave transmission and habitat creation.
Reef Height (h_R)	Total structure height, taken either at the toe or the centerline	Impacts freeboard but occasionally is used as a separate design parameter.
Gap Length (L_g)	Alongshore length of gaps between reef structures	The distance between reef segments influences wave transmission, sediment transport, and ecological connectivity.
Seaward Slope Angle (α)	Seaward slope angle of the structure	May impact wave steepness and location of breaking.
Rock Diameter (D_{50})	For rubble-mound structures, the median diameter of the armor stone	Since not all structures referenced in this section are rubble mound, sometimes this is modified or replaced with an effective diameter or similar value.
Offshore Significant Wave Height (H_s)	Deepwater significant wave height used for design and forcing	When considering irregular waves, this can often be replaced by the zero-th moment wave height (H_{m0}). It may also be replaced by the either the significant or zero-th moment wave height incident to the structure (H_i; typically after reflection is accounted for), depending on formulation.
Offshore Peak Wavelength (L_p)	Deepwater wavelength associated with peak wave period, used for design and forcing	This may occasionally be replaced by the wavelength associated with the mean spectral period (L_m), or taken closer to the structure (corresponding with H_i rather than H_s), depending on formulation.
Offshore Peak Wave Steepness (s_{op})	Deepwater wave steepness calculated as a ratio of offshore wave significant wave height to offshore peak wave period	While this is the most used form of wave steepness across the wave formula, it is important to note when the design waves are taken closer to the structure.
Iribarren Number (ξ)	Ratio of breakwater slope to the square root of offshore peak wave steepness	Indicator of the type of breaking expected, when followed by the subscript op, refers to the offshore peak value.

Godsey et al. (2026)

- Constructability: How (and why) do they recommend leaving latitude in construction?
- Monitoring and adaptive management: Is there a different in generalized monitoring and monitoring to inform adaptive management?

Table 24. Monitoring matrix requirements.

Performance Parameter	Monitoring Metric	Typical Trigger Condition	OMRR&R Action
Reef Elevation/Profile	Crest elevation, cross-section surveys	Elevation loss beyond design tolerance (e.g., >1 ft)	Add redistributed material to restore design
Reef Footprint/Width	Planform extent, bathymetry	Reduction in width or footprint due to spreading or loss	Regrade or place additional material to restore footprint
Structural Stability	Visual inspection, displacement mapping	Movement, settlement, or structural degradation	Repair, reinforce, or replace affected areas
Wave Attenuation Performance	Wave height measurements (if applicable)	Reduction below design performance target	Increase crest height, width, or adjust configuration
Sediment Deposition/Erosion	Accretion/erosion rates, burial depth	Excessive burial or scour affecting function	Sediment redistribution, elevation adjustment
Oyster Density/recruitment	Density counts, spat recruitment	Density below sustainability threshold	Add cultch, seed oysters, or modify substrate
Habitat Function	Species richness, structural complexity	Decline in habitat use or diversity	Supplement substrate or modify reef structure
Water Quality/Suitability	Salinity, dissolved oxygen, temperature	Conditions limiting oyster survival (site specific)	Evaluate need for intervention or adaptive management

Next Week

Assignment:
Turn in a draft of your trace-able account. These will be circulated to your classmates for review.

Papers:
~~Curran J.C., McKay S.K., Beagle J.R., Brauer E.J., Haring C.P., Thomas J., Lange J., Duhr D., VanBuecken N., Rose C., Bishop D., and Jensen S. 2025. Engineering practice guide for floodplain benching: A natural infrastructure approach for riverine systems. ERDC TR-25-15. U.S. Army Engineer Research and Development Center, Vicksburg, Mississippi.~~

Webb B., Dix B., Douglass S., Asam S., Cherry C., and Buhring B. 2019. Nature-based solutions for coastal highway resilience: An implementation guide. FHWA-HEP-19-042. Federal Highway Administration, Washington, DC. (***only Ch6 and Ch7***)